

Q1.

Solid, liquid and gas are three different states of matter.

- (a) Describe the difference between the solid and gas states, in terms of the arrangement and movement of their particles.

(4)

- (b) What is meant by 'specific latent heat of vaporisation'?

(2)

- (c) While a kettle boils, 0.018 kg of water changes to steam.

Calculate the amount of energy required for this change.

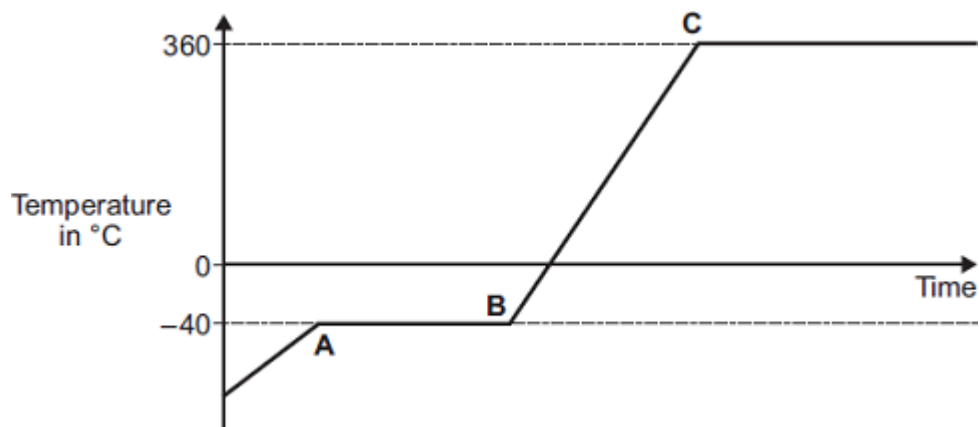
Specific latent heat of vaporisation of water = 2.3×10^6 J / kg.

Energy required = _____ J

(2)

- (d) The graph shows how temperature varies with time for a substance as it is heated.

The graph is **not** drawn to scale.



Explain what is happening to the substance in sections **AB** and **BC** of the graph.

Section **AB** _____

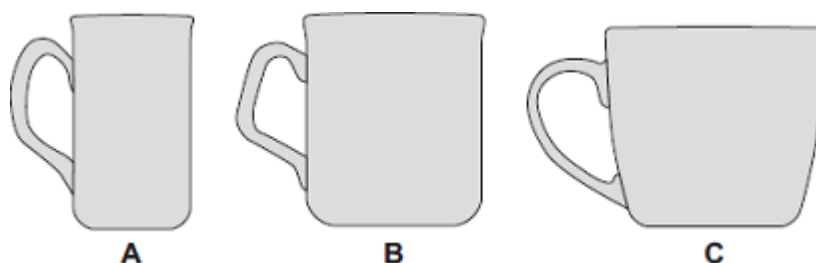
Section **BC** _____

(4)

(Total 12 marks)

Q2.

The diagram shows three cups **A**, **B** and **C**.



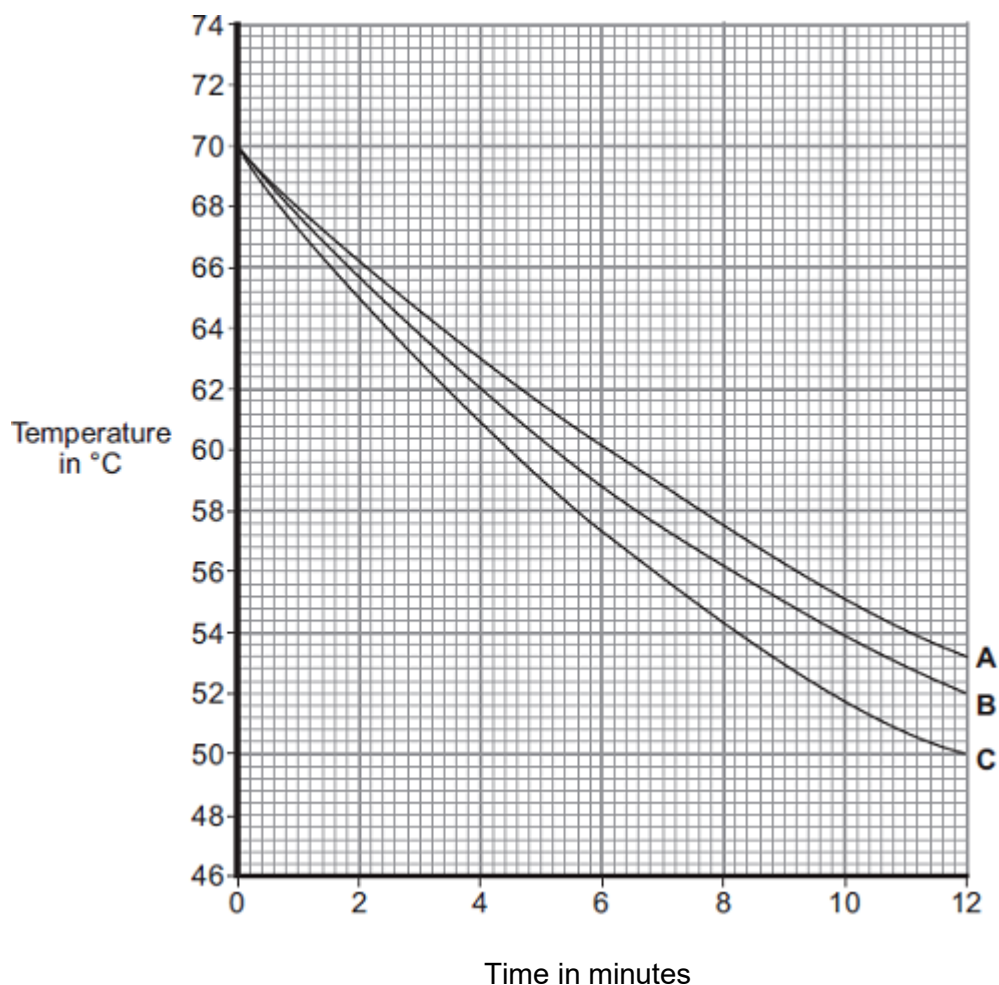
Energy is transferred from hot water in the cups to the surroundings.

(a) Some students investigated how the rate of cooling of water in a cup depends on

the surface area of the water in contact with the air.

They used cups **A**, **B** and **C**. They poured the same volume of hot water into each cup and recorded the temperature of the water at regular time intervals.

The results are shown on the graph.



- (i) What was the starting temperature of the water for each cup?

Starting temperature = _____ °C

(1)

- (ii) Calculate the temperature fall of the water in cup **B** in the first 9 minutes.

Temperature fall = _____ °C

(2)

- (iii) Which cup, **A**, **B** or **C**, has the greatest rate of cooling?



Using the graph, give a reason for your answer.

(2)

- (iv) The investigation was repeated using the bowl shown in the diagram.
The same starting temperature and volume of water were used.



Draw on the graph in part (b) another line to show the expected result.

(1)

- (v) After 4 hours, the temperature of the water in each of the cups and the bowl was 20°C.

Suggest why the temperature does **not** fall below 20°C.

(1)

- (b) (i) The mass of water in each cup is 200 g.

Calculate the energy, in joules, transferred from the water in a cup when the temperature of the water falls by 8°C.

Specific heat capacity of water = 4200 J / kg°C.

Energy transferred = _____ J

(3)

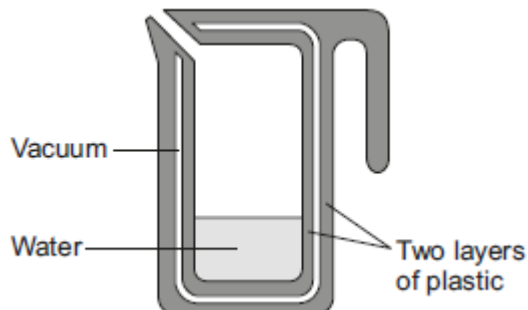
- (ii) Explain, in terms of particles, how evaporation causes the cooling of water.

(4)

Q3.

A new design for a kettle is made from two layers of plastic separated by a vacuum. After the water in the kettle has boiled, the water stays hot for at least 2 hours.

The new kettle is shown below.



- (a) The energy transferred from the water in the kettle to the surroundings in 2 hours is 46 200 J.

The mass of water in the kettle is 0.50 kg.

The specific heat capacity of water is 4200 J/kg °C.

The initial temperature of the water is 100 °C.

Calculate the temperature of the water in the kettle after 2 hours.

Temperature after 2 hours = _____ °C

(3)

- (b) Calculate the average power output from the water in the kettle to the surroundings in 2 hours.

Average power output = _____ W

(2)

(Total 5 marks)

Q4.

The photograph below shows a balloon filled with helium gas.



- (a) Which statements describe the movement of the gas particles in the balloon?

Tick (✓) **two** boxes.

The particles all move in a predictable way.

☐

The particles move at the same speed.

☐

The particles move in circular paths.

☐

The particles move in random directions.

☐

The particles move with a range of speeds.

☐

The particles vibrate about fixed positions.

☐

(2)

- (b) The pressure of the helium in the balloon is 100 000 Pa.

The volume of the balloon is 0.030 m³.

The balloon is compressed at a constant temperature causing the volume to decrease to 0.025 m³.

No helium leaves the balloon.

Calculate the new pressure in the balloon.

New pressure = _____ Pa

(4)

- (c) The temperature of the helium in the balloon was increased.

The mass and volume of helium in the balloon remained constant.

Explain why the pressure exerted by the helium inside the balloon would increase.

(4)

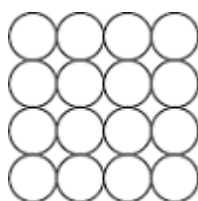
(Total 10 marks)

Q5.

According to kinetic theory, all matter is made up of small particles. The particles are constantly moving.

Diagram 1 shows how the particles may be arranged in a solid.

Diagram 1



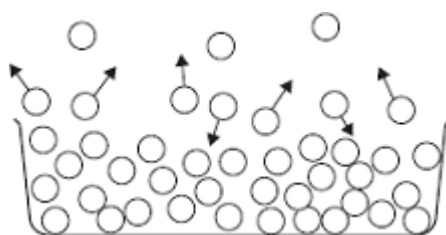
- (a) One kilogram of a gas has a much larger volume than one kilogram of a solid.

Use kinetic theory to explain why.

(4)

- (b) **Diagram 2** shows the particles in a liquid. The liquid is evaporating.

Diagram 2



- (i) How can you tell from **Diagram 2** that the liquid is evaporating?

(1)

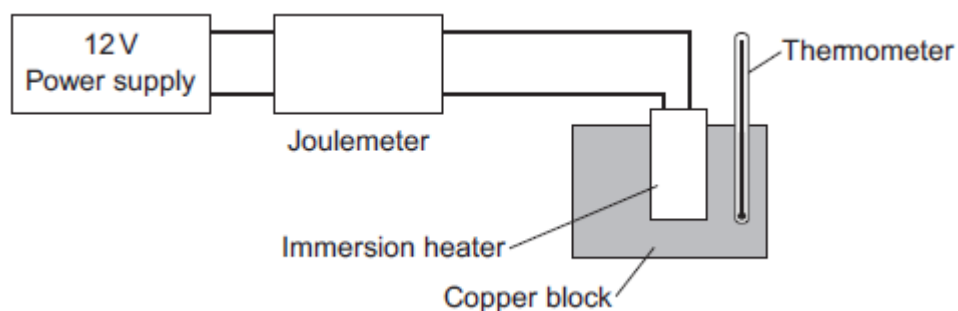
- (ii) The temperature of the liquid in the container decreases as the liquid evaporates.

Use kinetic theory to explain why.

Q6.

A student used the apparatus in **Figure 1** to obtain the data needed to calculate the specific heat capacity of copper.

Figure 1



The initial temperature of the copper block was measured.

The power supply was switched on.

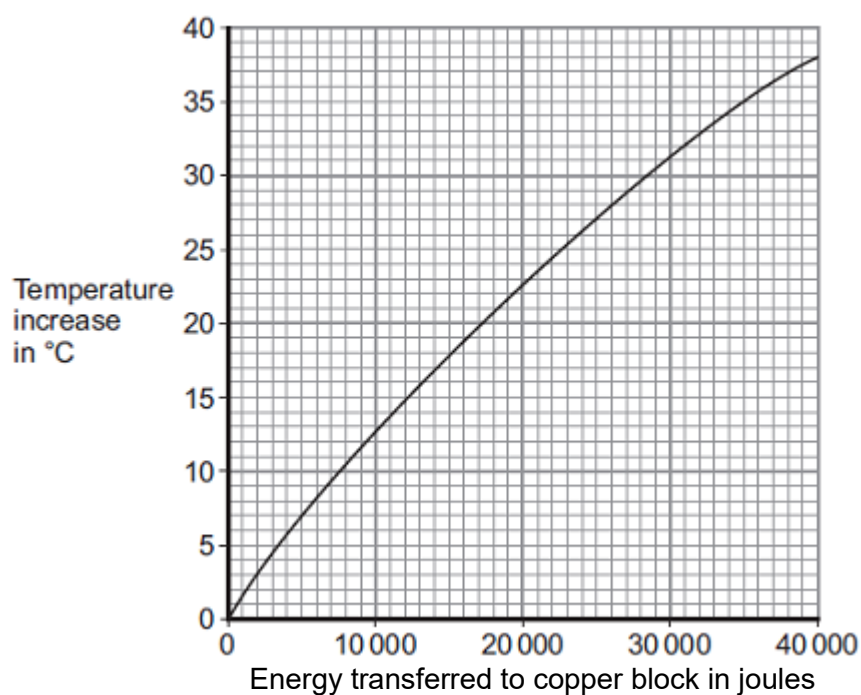
The energy transferred by the heater to the block was measured using the joulemeter.

The temperature of the block was recorded every minute.

The temperature increase was calculated.

Figure 2 shows the student's results.

Figure 2



- (a) Energy is transferred through the copper block.

What is the name of the process by which the energy is transferred?

Tick (✓) **one** box.

Conduction

☐

Convection

☐

Radiation

☐

(1)

- (b) Use **Figure 2** to determine how much energy was needed to increase the temperature of the copper block by 35 °C.

_____ joules

(1)

- (c) The copper block has a mass of 2 kg.

Use your answer to part (b) to calculate the value given by this experiment for the specific heat capacity of copper. Give the unit.

Specific heat capacity = _____

(3)

- (d) This experiment does **not** give the correct value for the specific heat of copper.

Suggest **one** reason why.

(1)

(Total 6 marks)

Mark schemes

Q1.

- (a) **solid**
particles vibrate about fixed positions 1
- closely packed
accept regular 1
- gas**
particles move randomly
accept particles move faster
accept freely for randomly 1
- far apart 1
- (b) amount of energy required to change the state of a substance from liquid to gas (vapour) 1
- unit mass / 1 kg
dependent on first marking point 1
- (c) 41000 **or** 4.1×10^4 (J)
accept
41400 or 4.14×10^4
correct substitution of
 $0.018 \times 2.3 \times 10^6$ gains 1 mark 2
- (d) **AB**
 changing state from solid to liquid / melting 1
- at steady temperature
dependent on first AB mark 1
- BC**
 temperature of liquid rises 1
- until it reaches boiling point
dependent on first BC mark 1

[12]

Q2.

- (a) (i) 70

		<i>accept $\pm$ half a square (69.8 to 70.2)</i>	1
(ii)	15	<i>accept 14.6 to 15.4 for 2 marks allow for 1 mark 70 – 55 ecf from (b)(i) $\pm$ half a square</i>	2
(iii)	C		1
		biggest drop in temperature during a given time <i>accept it has the steepest gradient this is a dependent</i>	1
(iv)	starting at 70 °C and below graph for C must be a curve up to at least 8 minutes		1
(v)	because 20 °C is room temperature <i>accept same temperature as surroundings</i>		1
(b)	(i) 6720 <i>correct answer with or without working gains 3 marks 6 720 000 gains 2 marks correct substitution of $E = 0.2 \times 4200 \times 8$ gains 2 marks correct substitution of $E = 200 \times 4200 \times 8$ gains 1 mark</i>		3
	(ii) the fastest particles have enough energy <i>accept molecules for particles</i>		1
	to escape from the surface of the water		1
	therefore the mean energy of the remaining particles decreases <i>accept speed for energy</i>		1
	the lower the mean energy of particles the lower the temperature (of the water) <i>accept speed for energy</i>		1
			[14]

Q3.

(a)	78 (°C)	<i>allow 2 marks for correct temperature change ie 22 °C allow 1 mark for correct substitution ie $46\,200 = 0.5 \times 4200 \times \theta$</i>
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or

$$\frac{46200}{0.5 \times 4200} = \theta$$

3

(b) 6.4 (W)

allow 2 marks for an answer that rounds to 6.4

allow 1 mark for correct substitution

ie $46\,200 = P \times 7200$

an answer of 23 000 or 23 100 or 385 gains 1 mark

2

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Q4.

(a) The particles move in random directions.

1

The particles move with a range of speeds.

1

(b) $100\,000 \times 0.030 = 3000$

1

$p \times 0.025 = 3000$

*allow a correct substitution using an incorrectly calculated value
using $pV = \text{constant}$*

1

$$p = \frac{3000}{0.025}$$

*allow a correct rearrangement using an incorrect value of the
constant*

1

$p = 120\,000 \text{ (Pa)}$

*allow a correct calculation using an incorrect value of the constant
allow correct substitution into $p_1V_1 = p_2V_2$ for first 2 marking points*

1

(c) particles would have a higher (mean) kinetic energy

*allow particles would have a higher (mean) speed
do not accept particles vibrate more*

1

(so) increased number of collisions with the walls of the balloon per second

allow greater frequency of collisions with the walls of the balloon

1

greater forces exerted in collisions (between particles and balloon walls)

allow greater rate of change of momentum (of particles)

1

greater force exerted on same area

allow description using $p = F/A$

Q5.

- (a) there are strong forces (of attraction) between the particles in a solid
accept molecules / atoms for particles throughout
accept bonds for forces

1

(holding) the particles close together
particles in a solid are less spread out is insufficient

1

or

(holding) the particles in a fixed pattern / positions

but in a gas the forces between the particles are negligible
accept very small / zero for negligible
accept bonds for forces

1

so the particles spread out (to fill their container)
accept particles are not close together
gas particles are not in a fixed position is insufficient

1

- (b) (i) particles are (shown) leaving (the liquid / container)
accept molecules / atoms for particles throughout
accept particles are escaping
particles are getting further apart is insufficient

1

- (ii) *accept molecules / atoms for particles throughout*
accept speed / velocity for energy throughout

particles with most energy leave the (surface of the) liquid
accept fastest particles leave the liquid

1

so the mean / average energy of the remaining particles goes down

1

and the lower the average energy (of the particles) the lower the temperature (of the liquid)

1

[8]

Q6.

- (a) conduction

1

- (b) 35 000

1

(c) 500

their (b) = 2 x c x 35 correctly calculated scores 2 marks
allow 1 mark for correct substitution,
ie 35000 = 2 x c x 35
or
their (b) = 2 x c x 35

2

J / kg°C

1

(d) energy lost to surroundings

or

energy needed to warm heater

accept there is no insulation (on the copper block)
*do **not** accept answers in terms of human error or poor*
results or defective equipment

1

[6]

Examiner reports

Q2.

- (a)
 - (i) Almost all students were able to read the starting value of temperature from a cooling curve.
 - (ii) Nearly all students correctly calculated the temperature fall from the cooling curve. Those who got it wrong gave the value of the temperature reached rather than the change in temperature.
 - (iii) The given graph showed the cooling curves for three cups of different cross-sectional areas. Students were asked which cup showed the greatest rate of cooling. Only half of the students were able to give a reason because they did not refer to temperature drop in a given time.
 - (iv) A diagram of a fourth container was given and students had to draw the expected cooling curve on the same axes. This was well done with four-fifths of students scoring full marks.
 - (v) Nearly all students recognised that the lowest temperature reached after four hours was also room temperature.
- (b)
 - (i) The calculation of energy transferred from the water, where the mass of water was given in grams, was correctly done by two-thirds of the students.
 - (ii) The explanation of evaporation causing the cooling of water was very poorly answered with half of the students scoring zero marks. Many students described convection and very few referred to the reduction in the mean energy of the particles when the most energetic had escaped from the surface of the water. Only a tenth of students scored three or four marks.

Q3.

- (a) This question discriminated well, a third of students scored all 3 marks. A third of students scored 2 marks for calculating the temperature change of 22°C . Surprisingly, many students thought the final temperature would be 122°C having added the 22°C to the initial temperature. These students still scored 2 marks for their calculation of temperature change provided working was shown. A surprising number of students failed to correctly subtract 22 from 100, believing the answer to be 88°C .

A number of students selected an incorrect equation. Some students scored 1 mark for the substitution only and then incorrectly rearranged to achieve an incorrect final answer. Students should be reminded that the substitution gains 1 mark, not the subsequent rearrangement.
- (b) Only one tenth of students scored 2 marks for the answer of 6.4 (W). Half the students scored 1 mark, with the most commonly seen answer of 23 100, which was achieved by dividing 46 200 by 2 (hours). A number of students attempted the conversion, but only got as far as minutes and ended with an answer of 385, which also scored 1 mark.

Q4.

- (a) 75% of students correctly identified two correct statements, while 23% of students

identified one.

- (b) 65% of students scored 4 marks. Students who understood that 'pressure $\times$ volume = a constant' were usually able to perform the calculation correctly. A few students made mistakes in copying down the quantities, eg missing off a 0 from, or adding an extra 0 to, the pressure. Many students who did not score marks thought that the relationship between the quantities was pressure/volume = a constant.
- (c) A good range of answers were seen and the question discriminated well between students. 1% of students scored all 4 marks, while 10% scored 3 marks, 25% scored 2 marks and 40% scored 1 mark. Most students realised that the answer involved the helium particles, and many referred correctly to the increase in kinetic energy or speed of these particles. However, few students were able to provide a coherent explanation beyond this. Whilst many referred to particle collisions, few indicated that the collisions were with the walls of the balloon and that the collisions would be more frequent. Many students seem to think that pressure is caused by particles colliding amongst themselves. A minority of answers went on to state that a greater force would be exerted on the balloon walls, but then failed to link this idea with the definition of pressure.

Q5.

- (a) Whilst many answers referred to the arrangement of particles in solids and gases, fewer responses referred to the forces between the particles.
- (b)
 - (i) Just under half of the students answered correctly in terms of particles leaving the container / liquid. Other answers seen just referred to the particles 'rising.'
 - (ii) Many answers referred to particles needing energy to escape, but often just stated 'gaining energy', without specifying that they had more energy than the particles that remained. Very few students gained the second marking point because they did not refer to "mean / average" energy. A fair number of students gained the third mark by linking a decrease in energy to the decrease in temperature.

Q6.

- (a) A very small amount of students did not identify conduction as the process by which energy is transferred through copper.
- (b) The majority of students answered correctly, of those who did not score the mark, the most common error was misreading the number on the x-axis (for a temperature increase of 35°C) as 30,500 instead of 35,000.
- (c) Around half of students scored two of the three marks available. This was usually for performing the calculation correctly, but failing to give the correct unit.
- (d) A very low proportion of students did not attempt this question, with less than a fifth scoring the mark. The most common incorrect answers referred to faulty apparatus, incorrect measurements or values not as stated in the question, e.g. the block was not 2kg.